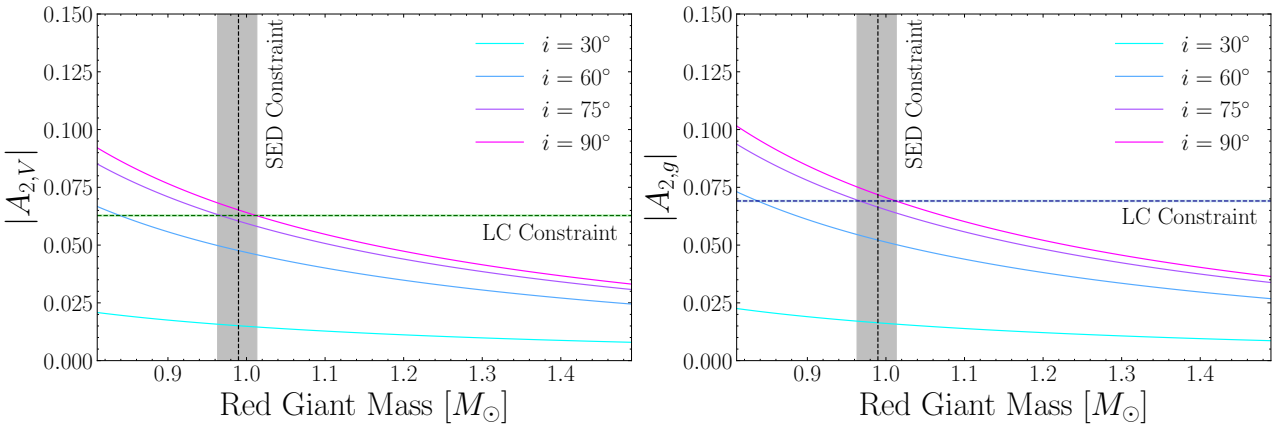

Figure 4. Dependence of the photometric amplitude of ellipsoidal variability (described by A_2) on the mass of the red giant and the orbital inclination in both the V-band (left panel) and g-band (right panel). The red giant radius is fixed to the best-fit value inferred from the SED. As the mass of the red giant increases, its Roche lobe filling factor drops, causing A_2 to decrease. In addition, as the orbital inclination decreases, the variability of the geometric cross-section becomes weaker, and A_2 tends to decrease. The respective constraints on A_2 from the observed light curve and the red giant mass from the observed SED then imply that the orbital inclination must be $> 75°$; the best-fit value is near $85°$.

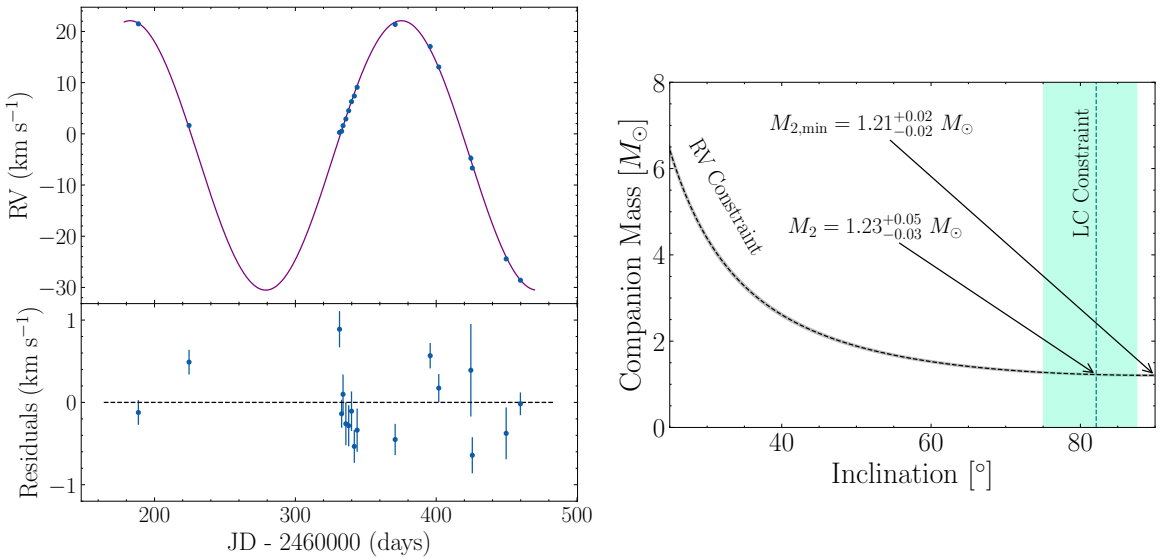

Figure 5. Left Panel: Radial velocity (RV) curve and RV residuals of the red giant in IGR J16194-2810. We measured RVs at 17 epochs over more than one orbit using FEROS spectra. The RV semi-amplitude of 26.3 ± 0.1 km s^{-1} implies a minimum companion mass of $1.21 \pm 0.02\ M_\odot$ (i.e. assuming an edge-on orbit), typical of a neutron star. Right Panel: Dynamical constraints on the companion mass. The RVs constrain the companion mass and inclination to the gray shaded region. The inclination constraint from the observed photometric variability amplitude (shaded green region) then implies $M_2 = 1.23^{+0.05}_{-0.03}\ M_\odot$.

$R_V = 3.1$ and a reddening of $E(B-V) = 0.45 \pm 0.02$ from the Green et al. (2019) 3D dust map. We use `pyphot`[2] to calculate synthetic photometry. The log-likelihood term for the SED data assumes Gaussian uncertainties:

$$\ln L_{\mathrm{SED}} = -\frac{1}{2} \sum_{\text{All Filters}} \left(\frac{m_{\mathrm{obs}} - m_{\mathrm{pred}}}{\sigma_m} \right)^2 \tag{5}$$

[2] https://mfouesneau.github.io/pyphot/

Our best-fit SED model (taken from the joint fit described in Section 5.6), along with its residuals, is presented in the left panel of Figure 3. In the right panel, we use the best-fit radius and effective temperature of the red giant (see Section 5.6) to compare it to MIST single-star evolutionary models (Choi et al. 2016), which imply an initial luminous star mass $M_* \approx 1.0\ M_\odot$ at an initial metallicity of $[\mathrm{Fe/H}] = -0.15$. As described in Section 5.6, this is the best-fit initial metallicity from our joint fit. The inferred age of the red giant is 9.9 ± 0.8 Gyr.

Table 3. All radial velocities (RVs) measured for IGR 16194-2810 using the FEROS spectra.

HJD UTC	RV (km s^{-1})
(1)	(2)
2460188.5062	21.49 ± 0.15
2460224.4984	1.62 ± 0.15
2460331.3500	0.27 ± 0.22
2460332.8462	0.51 ± 0.17
2460333.8497	1.59 ± 0.24
2460335.8577	2.90 ± 0.26
2460337.8621	4.51 ± 0.25
2460339.8638	6.28 ± 0.24
2460341.8519	7.39 ± 0.20
2460343.8616	9.09 ± 0.26
2460370.8610	21.37 ± 0.19
2460395.7393	17.05 ± 0.15
2460401.8136	13.05 ± 0.17
2460424.6068	-4.77 ± 0.56
2460425.6371	-6.69 ± 0.22
2460449.6892	-24.42 ± 0.32
2460459.7927	-28.60 ± 0.14

5.4. *Comparison to GALAH Spectra*

Deriving stellar parameters of cool giants is challenging, as the complex physics of red giant atmospheres involves broad molecular features blended with atomic lines, deviations from hydrostatic equilibrium (i.e. giant convective cells), and deviations from local thermal equilibrium (Bergemann et al. 2014).

To empirically estimate the atmospheric parameters of the red giant companion in IGR J16194-2810 and search for spectroscopic anomalies, we compare a high-resolution MIKE spectrum against the GALAH database (De Silva et al. 2015; Buder et al. 2021). Of our two MIKE spectra, we pick the one with lower resolution, but higher SNR. We degrade the MIKE spectrum to $R = 25,000$ and shift all spectra into their respective rest frames. Using the direct integration technique of Carvalho & Johns-Krull (2023), we apply a rotational broadening of $v \sin i = 15$ km s^{-1} to each GALAH spectrum (see Section 5.5). Then, we determine the closest match in pixel space by minimizing the total least-squares deviation between the spectra across all overlapping wavelength ranges that do not include telluric features. Based on the top ten closest matches, we find best-fit stellar parameters of $T_{\rm eff} = 3710 \pm 80$ K, $\log g = 0.71 \pm 0.14$, $[{\rm Fe/H}] = -0.7 \pm 0.4$, and $V_{\rm mic} = 2.20 \pm 0.11$ km s^{-1}.

To test the robustness of this approach, we compare our empirical parameters against those derived using the SED-based approach described by Reggiani et al. (2022). The SED-based effective temperature of $T_{\rm eff} = 3657 \pm 30$ and surface gravity of $\log g = 1.07 \pm 0.1$ are consistent with our results to within 2σ, while the SED-based metallicity of $[{\rm Fe/H}] = 0.3 \pm 0.12$ is discrepant.

We suspect that the GALAH metallicities for luminous giants are systematically underestimated: we find that the mean metallicity of all giants with $T_{\rm eff}$ and $\log g$ within 200 K and 0.5 dex of IGR 16194-2810 is $[{\rm Fe/H}] = -0.5$, which is significantly lower than the expected mean metallicity of $[{\rm Fe/H}] \sim -0.1$ in the solar neighborhood (e.g. Hayden et al. 2015). As we discuss in Section 5.6, the giant's true metallicity likely falls between the GALAH and SED-inferred values. We note that the *Gaia* XP metallicity inferred by Andrae et al. (2023) is $[{\rm M/H}] = -0.055$, in line with our expectations. The *Gaia* XP stellar parameters of $T_{\rm eff} = 3755$ K and $\log g = 0.93$ are also in 2σ agreement with our results.

We show a high-resolution optical MIKE spectrum of IGR J16194-2810, along with the most similar GALAH spectrum, in Figure 6. In the three panels, we compare the normalized and resolution-matched spectra of the two sources in the 2nd, 3rd, and 4th GALAH wavelength windows, respectively. The observed spectrum is typical of a M-type giant, with numerous absorption lines and molecular features evident. Most of the spectrum of IGR 16194-2810 is well-reproduced by its closest spectral doppelgangers. In general, the close match between the two spectra rules out a second luminous star, strong emission lines, or other unusual spectral features.

We also observe that the red giant's Hα absorption line is deeper and blueshifted relative to most similar sources observed by GALAH. We find some sources that have a similar Hα absorption line, including the best match displayed in Figure 6. This is most likely due to the outward expansion of the chromosphere in these red giants (see e.g. Mallik 1993) and unrelated to the companion.

5.5. *Measuring Projected Equatorial Rotational Velocity*

Measuring the projected equatorial rotation ($v \sin i$) of the red giant directly from observed high-resolution spectra provides an independent constraint on the stellar radius and orbital inclination inferred from the joint fitting described in Section 5.6. We begin by selecting a GALAH spectrum with a high similarity score (see Section 5.4) and a low $v \sin i$ as a template for a non-rotating red giant. For comparison, we select the highest-resolution MIKE spectra that we obtained. As before, we degrade the MIKE spectrum to $R = 25,000$ and shift all spectra into their respective rest frames. Then, for each order of the observed spectrum, we use Nelder-Mead optimization (Nelder & Mead 1965) to find the value of $v \sin i$ that minimizes the least-squares deviation between the spectral order and the rotationally broadened template over the relevant wavelength range. The rotational broadening is applied using the direct integration technique implemented by Carvalho & Johns-Krull (2023). Averaging across orders, we derive a projected equatorial rotational velocity of

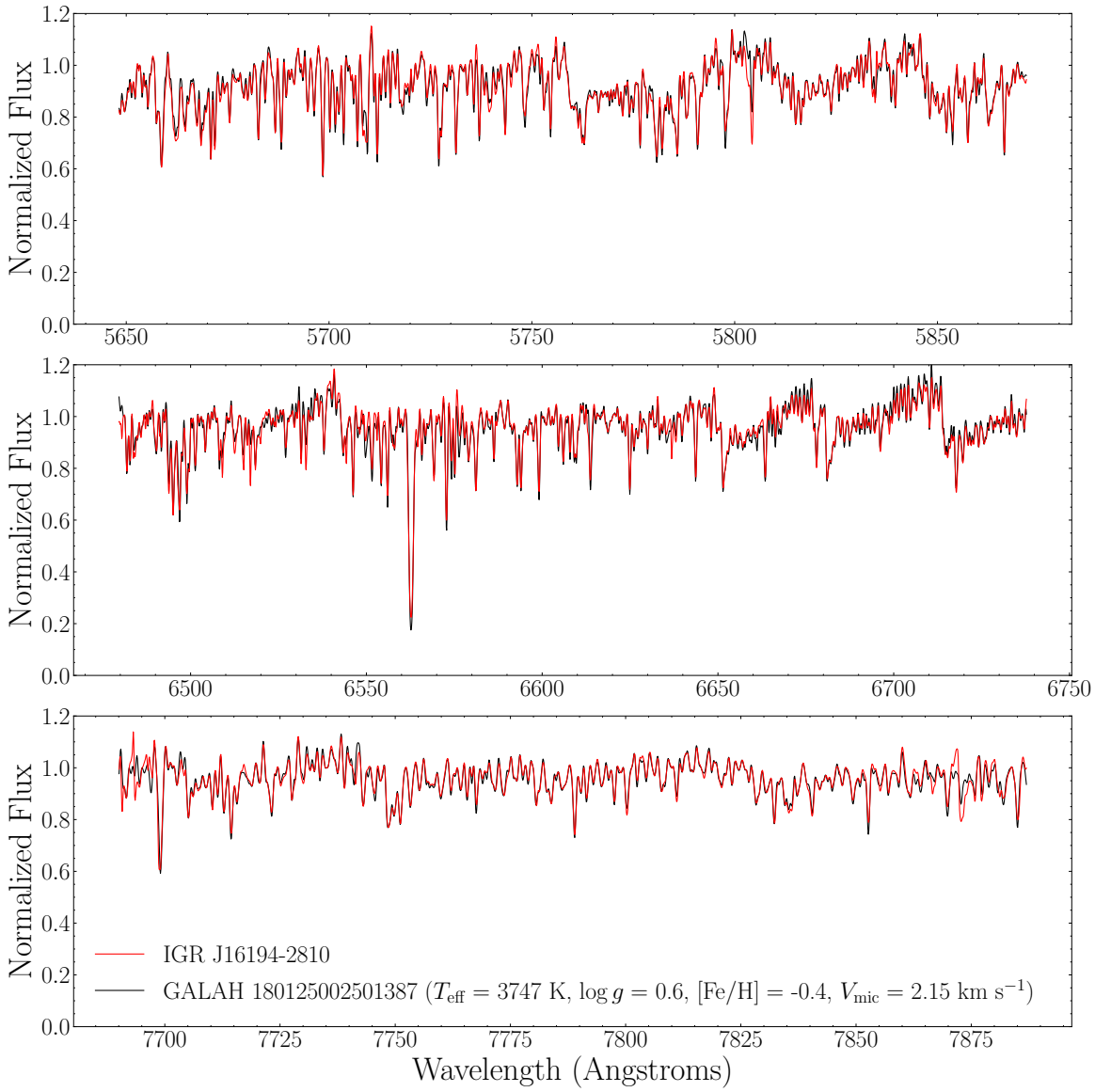

Figure 6. Cutouts of a high-resolution optical spectrum of IGR J16194-2810 obtained using the MIKE spectrograph on 02-04-2024. The Hα absorption line and molecular band features are evident. In each panel, the MIKE spectrum is compared against the corresponding region of the GALAH spectrum determined to be the closest match. While the two spectra appear to be quite similar, the inferred metallicity is in tension with the value derived for the red giant in IGR J16194-2810 from an SED-based approach.

$v \sin i = 17 \pm 2$ km s^{-1} for the red giant, where the error is derived based on the dispersion between orders.

5.6. *Joint Fitting with MCMC Techniques*

We simultaneously fit the light curves, RVs, stellar parameters, astrometry, and SED of IGR 16194-2810. The free parameters of our fit are the parallax, orbital period, orbital inclination, epoch of conjunction, center-of-mass RV, and NS mass, along with the red giant's age, initial mass, and initial metallicity. The observed ellipsoidal variability constrains the density and thus the present-day mass of the red giant, with some covariance with orbital inclination. The RVs constrain both the systemic velocity and the companion mass. When taken together, the light curves and RV curve constrain the orbital period and epoch of conjunction. The SED, together with the *Gaia* DR3 parallax, constrains the radius of the red giant. Given the radius and the effective temperature, MIST single-star evolutionary models constrain the red giant's age, initial mass, and initial metallicity. Lastly, the measured projected equatorial rotation constrains the inclination once the radius and orbital period are constrained by other information.

Unlike Hinkle et al. (2024), we do not use the lack of eclipses in the optical light curve of IGR 16194-2810 to place a limit on the inclination as a function of mass ratio. This is because — given the small physical size of the NS and ac-